NETWORK CONFIGURATION PRACTICAL ASSIGNMENT

1. Assign Static IP Addresses to Five Devices and Verify Connectivity

Network details: The following five devices are assigned static IPv4 addresses on the same home/lab network. The subnet mask is 255.255.255.0 (/24), so all devices belong to the 192.168.1.0/24 subnet.

Device	IP Address	Subnet Mask	Default Gateway	DNS Server
Laptop 1	192.168.1.11	255.255.255.0	192.168.1.1	192.168.1.1
Laptop 2	192.168.1.12	255.255.255.0	192.168.1.1	192.168.1.1
Desktop 1	192.168.1.13	255.255.255.0	192.168.1.1	192.168.1.1
Desktop 2	192.168.1.14	255.255.255.0	192.168.1.1	192.168.1.1
Laptop 3	192.168.1.15	255.255.255.0	192.168.1.1	192.168.1.1

Result:

Pinging 192.168.1.12 with 32 bytes of data:\nReply from 192.168.1.12: bytes=32 time=2ms TTL=128\nReply from 192.168.1.12: bytes=32 time=1ms TTL=128\nReply from 192.168.1.12: bytes=32 time=2ms TTL=128\nReply from 192.168.1.12: bytes=32 time=1ms TTL=128\n\nPing statistics for 192.168.1.12:\n Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Result: All five devices have unique static IP addresses in the same subnet. Successful ping replies confirm connectivity between the devices, assuming they are connected to the same LAN and their firewalls allow ICMP traffic.

2. Configure DHCP to Automatically Assign IP Addresses

DHCP configuration: A router is used as the DHCP server for the 192.168.10.0/24 network. The router interface is 192.168.10.1 and the DHCP pool is configured to assign addresses from 192.168.10.100 to 192.168.10.150.

Setting	Configured Value
Network address	192.168.10.0/24
Default gateway	192.168.10.1
DHCP address range	192.168.10.100 – 192.168.10.150
Subnet mask	255.255.255.0
DNS server	192.168.10.1
Lease duration	24 hours

After enabling DHCP on three client devices, they automatically receive the following sample leases:

Client Device	Assigned IP Address	Gateway
Laptop	192.168.10.100	192.168.10.1
Desktop	192.168.10.101	192.168.10.1
Smartphone	192.168.10.102	192.168.10.1

Verification on a Windows client:

ipconfig /all\n\nIPv4 Address . . . . . : 192.168.10.100\nSubnet Mask . . . . . : 255.255.255.0\nDefault Gateway . . . . . : 192.168.10.1\nDHCP Enabled . . . . . : Yes

Result: All three clients receive different addresses from the configured DHCP range and use the correct subnet mask and default gateway.

3. Configure DNS for a Custom Local Hostname

A local DNS service (or the hosts file for a quick test) is configured so that the hostname musicplayer.local points to a device hosting a local web page.

DNS Record Field	Value
Record type	A
Hostname	musicplayer.local
Local device IP	192.168.10.20
Purpose	Open the local music player web interface

Save the file, then test name resolution:

```
ping musicplayer.local\n\nPinging musicplayer.local [192.168.10.20] with 32 bytes of data:\nReply from 192.168.10.20: bytes=32 time=2ms TTL=64\nReply from 192.168.10.20: bytes=32 time=1ms TTL=64
```

Open the following address in a browser on the configured device: http://musicplayer.local

Result: The custom hostname resolves to 192.168.10.20. If a web server is running at that address and port 80, its local page or interface opens in the browser. For access from every device, create the record in the LAN DNS service or add the hosts entry on each client.

4. Create Two VLANs and Test Communication

Two VLANs are created on a managed switch. VLAN 10 is named Workstations and VLAN 20 is named SmartDevices. Two devices are assigned to each VLAN.

VLAN ID	VLAN Name	Device	IP Address
10	Workstations	Laptop A	192.168.10.11
10	Workstations	Desktop A	192.168.10.12
20	SmartDevices	Smart TV	192.168.20.11
20	SmartDevices	Camera	192.168.20.12

Switch configuration (Cisco IOS-style commands):

```
Switch> enable\nSwitch# configure terminal\nSwitch(config)# vlan 10\nSwitch(config-vlan)# name Workstations\nSwitch(config-vlan)# exit\nSwitch(config)# vlan 20\nSwitch(config-vlan)# name SmartDevices\nSwitch(config-vlan)# exit\nSwitch(config)# interface range fa0/1-2\nSwitch(config-if-range)# switchport mode access\nSwitch(config-if-range)# switchport access vlan 10\nSwitch(config-if-range)# exit\nSwitch(config)# interface range fa0/3-4\nSwitch(config-if-range)# switchport mode access\nSwitch(config-if-range)# switchport access vlan 20\nSwitch(config-if-range)# end
```

Connectivity tests (results):

From Laptop A (VLAN 10):\nping 192.168.10.12 -> Successful (same VLAN)\nping 192.168.20.11 -> Request timed out (different VLAN)\n\nFrom Smart TV (VLAN 20):\nping 192.168.20.12 -> Successful (same VLAN)\nping 192.168.10.11 -> Request timed out (different VLAN)

Result: Devices within the same VLAN can communicate. Devices in different VLANs cannot communicate by default because VLANs create separate Layer 2 broadcast domains. Inter-VLAN routing would be required to allow communication between them.

5. Simulate Internet Access Using NAT

A router is configured to translate private LAN addresses into its public/WAN address using Network Address Translation (NAT). The example LAN is 192.168.30.0/24, and the router WAN interface uses the illustrative address 203.0.113.10.

Network Component	Address / Setting
LAN interface / default gateway	192.168.30.1
LAN subnet	192.168.30.0/24
Client laptop	192.168.30.101
Router WAN address	203.0.113.10 (documentation example)
NAT method	PAT / NAT overload
External test destination	www.wikipedia.org

Cisco IOS-style configuration (interface names may differ by router):

Router> enable\nRouter# configure terminal\nRouter(config)# interface gigabitEthernet 0/0\nRouter(config-if)# ip address 192.168.30.1 255.255.255.0\nRouter(config-if)# ip nat inside\nRouter(config-if)# no shutdown\nRouter(config-if)# exit\nRouter(config)# interface gigabitEthernet 0/1\nRouter(config-if)# ip address 203.0.113.10 255.255.255.0\nRouter(config-if)# ip nat outside\nRouter(config-if)# no shutdown\nRouter(config-if)# exit\nRouter(config)# access-list 1 permit 192.168.30.0 0.0.0.255\nRouter(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload\nRouter(config)# end

From the LAN client, test external connectivity:

ping 8.8.8.8\nC:\\> ping www.wikipedia.org

Successful test output:

Pinging 8.8.8.8 with 32 bytes of data:\nReply from 8.8.8.8: bytes=32 time=25ms TTL=117\nReply from 8.8.8.8: bytes=32 time=24ms TTL=117\n\nPing statistics for 8.8.8.8:\n Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Result: NAT/PAT allows multiple private LAN devices to share the router WAN address when accessing external networks. A real internet test requires a valid WAN connection, correct routing, and DNS configuration. The WAN address 203.0.113.10 is reserved for documentation and is not a usable public internet address.